XML is Semi-structured Data

- Attributes with different types in different objects

```
<person> <name>  <first> John </first>
                 <last> Smith </last>
         </name>
         <phone>1234</phone>
</person>
```

Structured name !

- Nested collections (no 1NF)
- Heterogeneous collections:
  - <db> contains both <book>s and <publisher>s

Document Type Definitions DTD

- part of the original XML specification
- an XML document may have a DTD
- XML document:

  Well-formed = if tags are correctly closed

  Valid = if it has a DTD and conforms to it
- validation is useful in data exchange

DTD

Goals:

- Define what tags and attributes are allowed
- Define how they are nested
- Define how they are ordered

Superseded by XML Schema

- Very complex: DTDs still used widely

Very Simple DTD

```
<!DOCTYPE company [
  <!ELEMENT company   ((person|product)*)>
  <!ELEMENT person  (ssn, name, office, phone?)>
  <!ELEMENT ssn        (#PCDATA)>
  <!ELEMENT name     (#PCDATA)>
  <!ELEMENT office     (#PCDATA)>
  <!ELEMENT phone    (#PCDATA)>
  <!ELEMENT product  (pid, name, description?)>
  <!ELEMENT pid    (#PCDATA)>
  <!ELEMENT description    (#PCDATA)>
]>
```

Very Simple DTD

Example of valid XML document:

```
<company>
   <person>  <ssn> 123456789 </ssn>
             <name> John </name>
             <office> B432 </office>
             <phone> 1234 </phone>
   </person>
   <person>  <ssn> 987654321 </ssn>
             <name> Jim </name>
             <office> B123 </office>
   </person>
   <product> ...  </product>
   ...
</company>
```

DTD: Regular Expressions

DTD

sequence

```
<!ELEMENT name
                (firstName, lastName))>
```

XML

```
<name>
   <firstName>  . . . . . </firstName>
   <lastName>  . . . . . </lastName>
</name>
```

optional

```
<!ELEMENT name  (firstName?, lastName))>
```

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```
<!ELEMENT person   (name, phone*))>
```

```
<person>
   <name>  . . . . . </name>
   <phone>  . . . . . </phone>
   <phone>  . . . . . </phone>
   <phone>  . . . . . </phone>
   . . . . . .
</person>
```

alternation

```
<!ELEMENT person   (name, (phone|email)))>
```

DTD v.s. XML Schemas

DTD:

```
<!ELEMENT paper (title,author*,year, (journal|conference))>
```

XML Schema:

```
<xs:element name="paper" type="paperType"/>
<xs:complexType name="paperType">
    <xs:sequence>
        <xs:element name="title" type="xs:string"/>
        <xs:element name="author" minOccurs="0"/>
        <xs:element name="year"/>
        <xs: choice> < xs:element name="journal"/>
                        <xs:element name="conference"/>
        </xs:choice>
    </xs:sequence>
</xs:element>
```

Example

A valid XML Document:

```
<paper>
  <title> The Essence of XML </title>
  <author> Simeon</author>
  <author> Wadler</author>
  <year>2003</year>
  <conference> POPL</conference>
</paper>
```